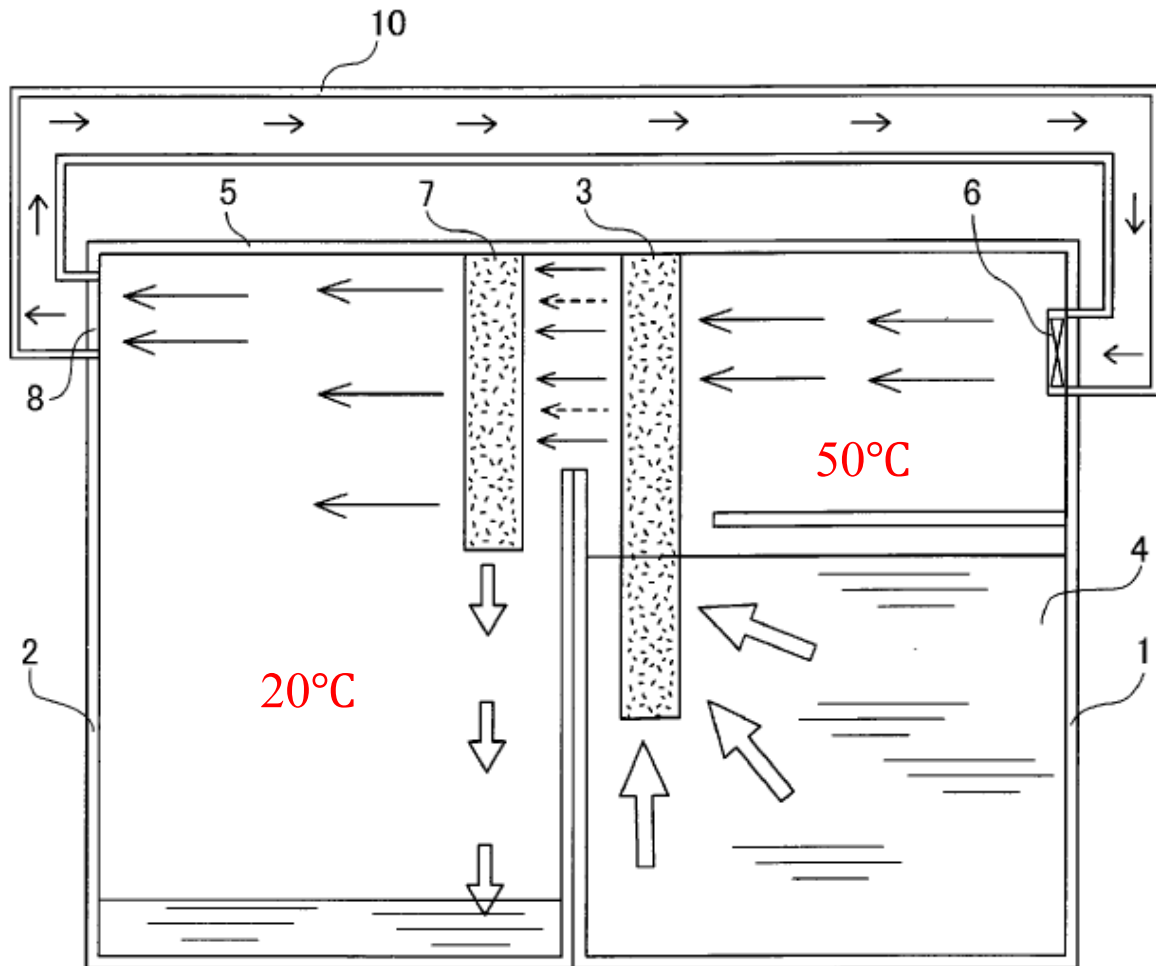


Amount of saturated water vapor:

50°C 82.8g/m³

20°C 17.2g/m³

difference:

50°C—20°C

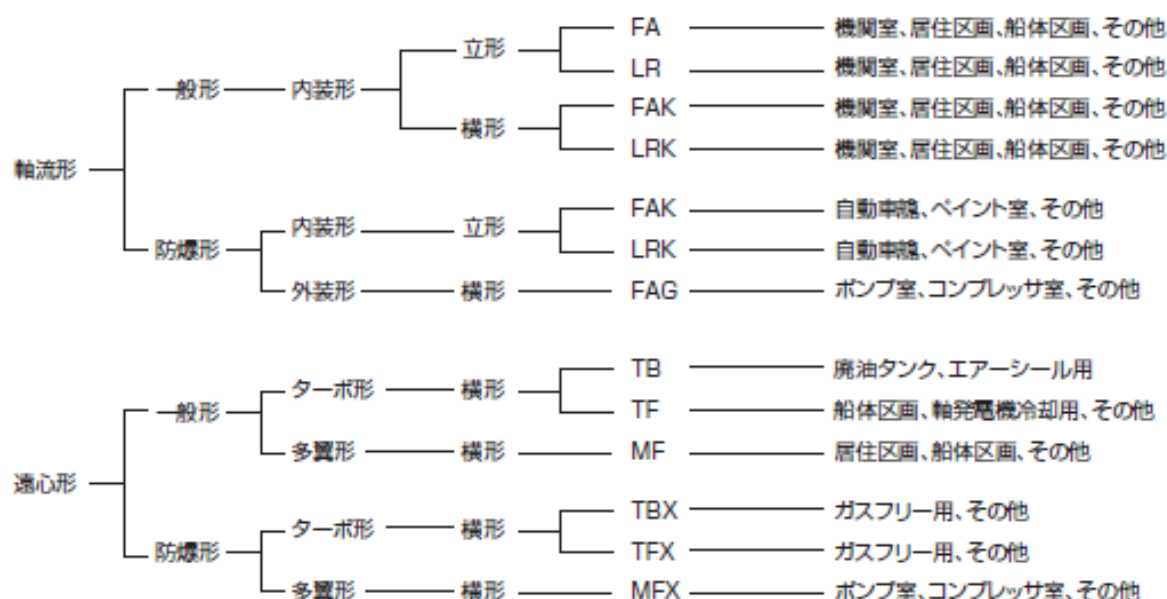
82.8—17.2=65.6g/m³

Blower: TAIYO Model FA-B 3,500m³/min.

65.6 × 3,500=229,600g/min.

229,000 × 60min. × 24h=330,624,000g/Day

About 330ton/Day



注1. 軸流形ファンのケーシングを2つ糊した場合のモデル名は、-Bを付ける。(例FA-B)

注2. 軸流ファンに笠、ダンパーを装着した場合のモデル名は、-Dを付ける。(例FA-D)

代表モデル

モデルFA-B

立軸流電動機内装形

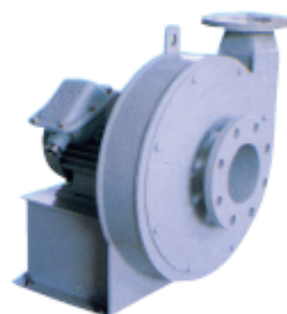
- 風量: ~3500m³/min
- 静圧: ~0.98kPa
- 出力: ~90kW



モデルTB

遠心式ターボ形

- 風量: ~200m³/min
- 静圧: ~14.7kPa
- 出力: ~90kW



モデルFAX-B

立軸流電動機内装形
電動機耐圧防爆形
ノンスパーク形

- 風量: ~3500m³/min
- 静圧: ~0.98kPa
- 出力: ~90kW



モデルTFX

遠心式ターボ形
ノンスパーク形

- 風量: ~1000m³/min
- 静圧: ~4.9kPa
- 出力: ~160kW



モデルFAG

横軸流電動機外装形
ノンスパーク形

- 風量: ~2000m³/min
- 静圧: ~0.98kPa
- 出力: ~90kW

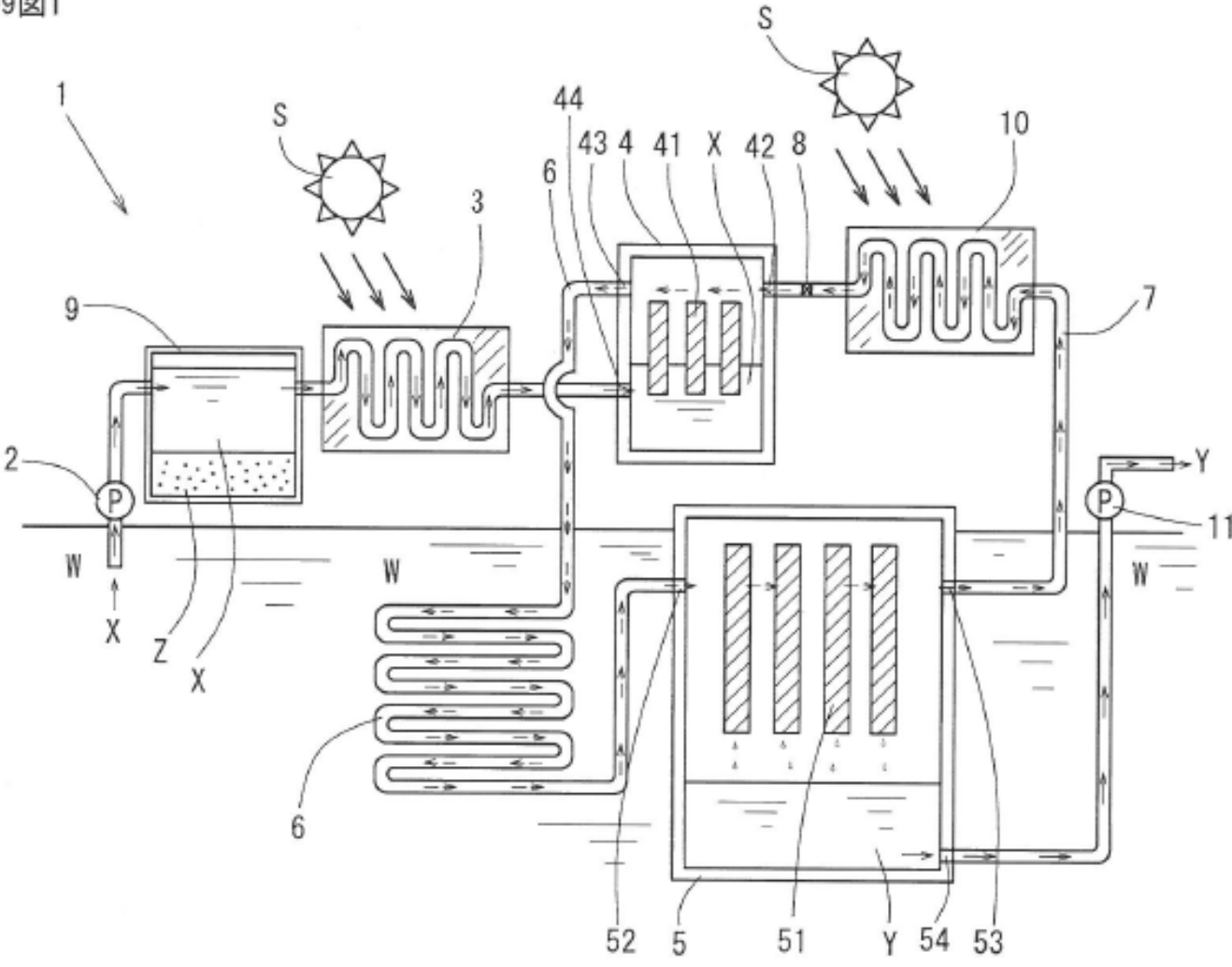


モデルMFX

遠心式多翼形
ノンスパーク形

- 風量: ~600m³/min
- 静圧: ~0.98kPa
- 出力: ~37kW





Evidence and data

Related intellectual property rights

Patent Application No. 2014-265431(JPN), 2014/12/26

Patent No. 6391698 (JPN), 2018/3/31 Registered

PCT / JP2016 / 57553 WO2016143848 A1 (international)

【Investigation result of "International search report system in PCT application" by Patent Office of Japan】

WO2016143848 A1 “Fresh water generating apparatus”

International search report: 6 preliminary documents have been reported, and all documents are [A] = “document defining the general state of the art which is not considered to be of particular relevance”

Opinion written by the International Searching Authority: "The inventions according to claims 1 to 8 are not described in any documents described in the international search report, nor are they obvious to those skilled in the art from the description of the above documents "

Seawater desalination Test

Test items	Unit	Raw water	Treated water	Residual water
Hydrogen ion concentration	pH	7.2	5.3	7.2
Amount of floating substance	mg/L	2	3	270
Sodium	mg/L	15,000	26	66,000
Chloride ion	mg/L	31,000	31	140,000
Total Water Hardness	mg/L	170	4	650

(12) 特許協力条約に基づいて公開された国際出願

(19) 世界知的所有権機関
国際事務局

(43) 国際公開日
2016年9月15日(15.09.2016)



(10) 国際公開番号

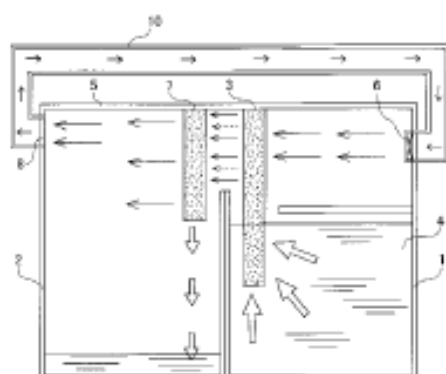
WO 2016/143848 A1

- (51) 国際特許分類:
C02F 1/04 (2006.01) B01D 1/00 (2006.01)
- (21) 国際出願番号: PCT/JP2016/057553
- (22) 国際出願日: 2016年3月10日(10.03.2016)
- (25) 国際出願の言語: 日本語
- (26) 国際公開の言語: 日本語
- (30) 優先権データ:
特願 2015-047132 2015年3月10日(10.03.2015) JP
- (71) 出願人: 株式会社ワンワールド (ONEWORLD CORPORATION) [JP/JP]; 〒5440015 大阪府大阪市生野区箕面南四丁目7番24号 Osaka (JP).
- (72) 発明者: および
- (71) 出願人: 伊藤 智章 (ITO Tomoaki) [JP/JP]; 〒5360014 大阪府大阪市城東区鳴野西四丁目2番3号104 Osaka (JP).
- (74) 代理人: 中井 信宏 (NAKAI Nobuhiro); 〒5400031 大阪府大阪市中央区北浜東1-1-2 千歳第一ビル4階 Osaka (JP).
- (81) 指定国 (表示のない限り、全ての種類の国内保護が可能): AE, AG, AL, AM, AO, AT, AU, AZ, BA, BB, BG, BH, BN, BR, BW, BY, BZ, CA, CH, CL, CN, CO, CR, CU, CZ, DE, DK, DM, DO, DZ, EC, EE, EG, ES, FI, GB, GD, GE, GH, GM, GT, HN, HR, HU, ID, IL, IN, IR, IS, JP, KE, KG, KN, KP, KR, KZ, LA, LC, LK, LR, LS, LU, LY, MA, MD, ME, MG, MK, MN, MW, MX, MY, MZ, NA, NG, NI, NO, NZ, OM, PA, PE, PG, PH, PL, PT, QA, RO, RS, RU, RW, SA, SC, SD, SE, SG, SK, SL, SM, ST, SV, SY, TH, TJ, TM, TN, TR, TT, TZ, UA, UG, US, UZ, VC, VN, ZA, ZM, ZW.
- (84) 指定国 (表示のない限り、全ての種類の広域保護が可能): ARIPO (BW, GH, GM, KE, LR, LS, MW, MZ, NA, RW, SD, SL, ST, SZ, TZ, UG, ZM, ZW), ユーラシア (AM, AZ, BY, KG, KZ, RU, TJ, TM), ヨーロッパ (AL, AT, BE, BG, CH, CY, CZ, DE, DK, EE, ES, FI, FR, GB, GR, HR, HU, IE, IS, IT, LT, LU, LV, MC, MK, MT, NL, NO, PL, PT, RO, RS, SE, SI, SK, SM, TR), OAPI (BF, BJ, CF, CG, CI, CM, GA, GN, GQ, GW, KM, ML, MR, NE, SN, TD, TG).
- 添付公開書類:
— 国際調査報告 (条約第21条(3))

(54) Title: FRESH WATER-GENERATING APPARATUS

(54) 発明の名称: 真水生成装置

(図1)



← 水 AA
← 送風 BB
← 水蒸気 CC

AA Water
BB Air flow
CC Water vapor

(57) Abstract: [Problem] To provide a fresh water-generating apparatus that does not require large amounts of heating energy as in the past, also does not require maintenance such as replacement of reverse osmosis membranes, and is simpler and inexpensive. [Solution] A fresh water-generating apparatus is configured from: a first porous water-absorbing substrate 3, each hole diameter of the porous structure that is immersed in a stored liquid 4 stored in a first storage tank 1 being formed in the size of a water molecule; a blower 6 for blowing air on the upper end of the first porous water-absorbing substrate 3; a second porous water-absorbing substrate 7 for catching water vapor, which is vaporized from the upper end of the first porous water-absorbing substrate 3 by the air flow from the blower 6 and is blown out, on the front surface thereof; a storage tank 2 for storing water that condenses inside the second porous water-absorbing substrate 7 and drips from the lower part of the second porous water-absorbing substrate 7; and a guide pipe 8 for connecting air that is sent out and discharged from the second porous water-absorbing substrate 7 to the air intake side of the blower 6.

(57) 要約: 【課題】従来のような多量な熱エネルギーを必要とせず、逆浸透膜のような交換メンテも必要としない、より簡易でコストもかからない真水生成装置を提供する。【解決手段】第一の貯蔵槽 1 に貯蔵された貯蔵液 4 に浸漬された多孔質構造の各孔径が水分子の大きさに形成された第一の多孔質吸水基板 3 と、第一の多孔質吸水基板 3 の上端部側に送風する送風機 4 と、送風機 4 による送風によって第一の多孔質吸水基板 3 の上部側から気化し吹き出される水蒸気をその前面側で受け止める第二の多孔質吸水基板 7 と、第二の多孔質吸水基板 7 内で凝結し、第二の多孔質吸水基板 7 の下部側から落滴する水を貯蔵する貯蔵槽 2 と、第二の多孔質吸水基板 7 から送り出され排気される空気を送風機 6 の吸気側へ連通させる導通管 8 とから構成する。

INTERNATIONAL SEARCH REPORT

International application No.

PCT/JP2016/057553

A. CLASSIFICATION OF SUBJECT MATTER

C02F1/04(2006.01)1, B01D1/00(2006.01)1

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

C02F1/04, B01D1/00

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Jitsuyo Shinan Koho 1922-1996 Jitsuyo Shinan Toroku Koho 1996-2016

Kokai Jitsuyo Shinan Koho 1971-2016 Toroku Jitsuyo Shinan Koho 1994-2016

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

DWPI(Thomson Innovation)

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
A	WO 2004/067451 A1 (SCHOBBER, Rudolf), 12 August 2004 (12.08.2004), claims; page 4, line 13 to page 6, line 17; drawings & IT MI20040041 U1	1-8
A	JP 2011-31157 A (Yugen Kaisha Re Earth), 17 February 2011 (17.02.2011), claims; paragraphs [0016] to [0032]; drawings (Family: none)	1-8
A	JP 2-191586 A (Shunpei KAWANAMI), 27 July 1990 (27.07.1990), claims; page 2, lower right column, line 19 to page 4, upper left column, line 6; drawings (Family: none)	1-8

☒ Further documents are listed in the continuation of Box C.☐ See patent family annex.

* Special categories of cited documents:

"A" document defining the general state of the art which is not considered to be of particular relevance

"E" earlier application or patent but published on or after the international filing date

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"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&" document member of the same patent family

Date of the actual completion of the international search

11 April 2016 (11.04.16)

Date of mailing of the international search report

19 April 2016 (19.04.16)

Name and mailing address of the ISA/

Japan Patent Office

3-4-3, Kasumigaseki, Chiyoda-ku,

Tokyo 100-8915, Japan

Authorized officer

Telephone No.

INTERNATIONAL SEARCH REPORT

International application No.

PCT/JP2016/057553

C (Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
A	GB 2330779 A (FILEWOOD Alan Roy), 05 May 1999 (05.05.1999), claims; page 9, line 9 to page 11, line 27; drawings (Family: none)	1-8
A	GB 2400603 A (FILEWOOD Alan Roy), 20 October 2004 (20.10.2004), claims; page 9, line 19 to page 13, line 5; drawings & GB 2400603 B	1-8
A	JP 2012-40454 A (Kabushiki Kaisha M Hikari Energy Kaihatsu Kenkyusho), 01 March 2012 (01.03.2012), claims; paragraphs [0022] to [0024]; drawings (Family: none)	1-8